EXPERIMENT 22: DIETING SYSTEM BASED ON DISEASE

Aim

To write a Prolog program that suggests a suitable diet based on a person's disease using facts and rules.

Algorithm

1. Start.
2. Define diseases and their recommended foods.
3. Define foods to avoid.
4. Accept a disease as a query.
5. Search the database for suitable food.
6. Display the recommended diet.
7. Stop.

Source Code

```
% Diet Recommendation System
diet(diabetes, vegetables).
diet(diabetes, whole_grains).
diet(diabetes, salads).

diet(hypertension, fruits).
diet(hypertension, vegetables).
diet(hypertension, low_salt_food).

diet(obesity, vegetables).
diet(obesity, fruits).
diet(obesity, low_calorie_food).

avoid(diabetes, sugary_food).
avoid(hypertension, salty_food).
```

```
avoid(obesity, fast_food).


recommend(Disease, Food) :-
    diet(Disease, Food).
```

Queries and Output

?- recommend(diabetes, Food).

Food = vegetables ;

Food = whole_grains ;

Food = salads.

?- recommend(hypertension, Food).

Food = fruits ;

Food = vegetables ;

Food = low_salt_food.

OUTPUT

```
?- [diet].
% diet.pl compiled 0.00 sec, 14 clauses

?- recommend(diabetes, Food).
Food = vegetables ;
Food = whole_grains ;
Food = salads.
true.

?- recommend(hypertension, Food).
Food = fruits ;
Food = vegetables ;
Food = low_salt_food.
true.

?-
```

Result

The Prolog program successfully suggests diet items based on the given disease using a simple rule-based knowledge base.